One easy instance of when $AB=BA$ is if $B=A$.

· I don't have any method to figure out ~~a~~ a B s.t. $B \neq A$ and $AB=BA$. □

12. Let $A=\begin{bmatrix} 3 & -6 \\ -1 & 2 \end{bmatrix}$. Construct a 2×2 matrix B such that AB is the zero matrix. Use two different nonzero columns for B.

$\boxed{\text{Solution:}}$ Let $B=\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$. Atleast one of b_{11} and b_{21} is non-zero.

Atleast one of b_{12} and b_{22} is non-zero.

$AB=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & -6 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\Rightarrow [Ab_1 \ Ab_2] = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (By defn. of matrix multiplication)

$Ab_1 = \begin{bmatrix} 3 & -6 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix} = b_{11}\begin{bmatrix} 3 \\ -1 \end{bmatrix} + b_{21}\begin{bmatrix} -6 \\ 2 \end{bmatrix}$ (By defn. of Ax)

$= \begin{bmatrix} 3b_{11} \\ -b_{11} \end{bmatrix} + \begin{bmatrix} -6b_{21} \\ 2b_{21} \end{bmatrix}$ (By defn. of scalar multiple)

$= \begin{bmatrix} 3b_{11}-6b_{21} \\ -b_{11}+2b_{21} \end{bmatrix}$ (By defn. of vector sum)

$Ab_2 = \begin{bmatrix} 3 & -6 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} b_{12} \\ b_{22} \end{bmatrix} = \begin{bmatrix} 3b_{12}-6b_{22} \\ -b_{12}+2b_{22} \end{bmatrix}$ (Similar calculation as Ab_1)

$\therefore AB = \begin{bmatrix} 3b_{11}-6b_{21} & 3b_{12}-6b_{22} \\ -b_{11}+2b_{21} & -b_{12}+2b_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\therefore 3b_{11}-6b_{21}=0 \Rightarrow b_{11}=2b_{21}$ $\quad -b_{11}+2b_{21}=0 \Rightarrow b_{11}=2b_{21}$

$3b_{12}-6b_{22}=0 \Rightarrow b_{12}=2b_{22}$ $\quad -b_{12}+2b_{22}=0 \Rightarrow b_{12}=2b_{22}$.

Let $b_{21}=1 \ \therefore b_{11}=2$ $\qquad B=\begin{bmatrix} 2 & 10 \\ 1 & 5 \end{bmatrix}$

Let $b_{12}=10 \ \therefore b_{22}=5$

13. Let $r_1, \dots, r_p$ be vectors in $\mathbb{R}^n$, and let Q be an $m\times n$ matrix. Write the matrix $[Qr_1, \dots, Qr_p]$ as a product of two matrices (neither of which is an identity matrix).